REPORT 008: NKEDITOR4 — EPOCH 008: 8OPPERC PERFORMANCE DOMAIN MODEL & DUAL CONTROLLER MAPPER

2026-11-04 00:49:40 | 8opPerc Percussion Rig Domain Model in Rust & Node.js, Native nanoKONTROL2 Mapping, and MIDImix Emulation

Operator: Antigravity Agent · Follows: Report 007

Prompt

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Implement the 8opPerc performance domain model in Rust (`src/op8_model.rs`) and
↪  Node.js (`js/op8-nanokontrol2-mapper.js`), referencing
↪  `docs/NKEDITOR-RUST-PORT-SPEC.md` and
↪  `/home/renick/claude/8operc-konduktiva/src/pulsed-live-rig.js`. Model all 8
↪  percussion voice lanes (`FAMILY_ORDER`), per-lane density multipliers (`0.1` to
↪  `5.0`), per-lane macro/gain sliders, per-lane Mute and new Solo states, per-lane
↪  active and queued timbre names, global density multiplier (`0.1` to `5.0`),
↪  global rhythmic feel scale (`swing` and `humanize` `0.0` to `1.0` with exact
↪  midpoint `CC 64 = 0.5`), 4,096-entry form bank state, and 4,096-entry timbre bank
↪  state. Implement both (1) Native nanoKONTROL2 Performance Mapping (`Knobs CC
↪  16-23` = voice density, `Sliders CC 0-7` = voice gain/macro, `Solo CC 32-39` =
↪  voice solo, `Mute CC 48-55` = voice mute, `Rec CC 64-71` = lane timbre reroll,
↪  `Track L/R CC 58/59` = form prev/next, `Cycle CC 46` = full timbre set, `Marker
↪  Set CC 60` = atomic preset pool capture, `Marker L/R CC 61/62` = global density
↪  step, `Rewind/Forward CC 43/44` = global feel step, `Play/Stop/Rec CC 41/42/45` =
↪  transport) and (2) MIDImix Emulation Mode translating nanoKONTROL2 into factory
↪  MIDImix Notes/CCs. Add Rust and Node.js tests (with `CARGO_BUILD_JOBS=2`), git
↪  commit all changes, and compile `reports/report-008.html` from
↪  `reports/report-008.md` via
↪  `/home/renick/claude/html-report-generator/build-report.sh --pdf` and commit.
```

TL;DR

Implemented the 8opPerc performance domain model in Rust (src/op8_model.rs) and Node.js (js/op8-nanokontrol2-mapper.js), modeling all 8 percussion voice lanes (FAMILY_ORDER), exponential density ladders (0.1 to 5.0), exact midpoint feel scaling (CC 64 = 0.5), Solo/Mute priority audibility, 4,096-entry form and timbre banks, Native nanoKONTROL2 Performance Mapping, and MIDImix Emulation Mode. All 38 Rust tests and 9 Node.js tests pass (47 total).

Plain-English Overview: What Changed and Why It Matters

In live electronic percussion performance with `8opPerc` (an 8-operator FM percussion synthesizer) and `Konduktiva` (a musical scheduling environment), a performer controls eight simultaneous drum voices—such as kicks, snares, hats, toms, and cymbals—while navigating thousands of pregenerated musical forms and sound palettes (timbres). Previously, the live rig (`pulsed-live-rig.js`) was hardwired only for the Akai `MIDI Mix` controller, which lacked dedicated transport buttons and relied on an awkward 32-message ”Send All” burst to save a preset snapshot.

In Epoch 8, we built the complete `8opPerc` performance brain in both Rust (`src/op8_model.rs`) and Node.js (`js/op8-nanokontrol2-mapper.js`):

- **Eight Percussion Voice Lanes with Solo Priority**: Each of the 8 drum lanes tracks its own note density multiplier (`0.1x` sparse to `5.0x` dense), volume/macro slider (`0..127`), Mute button, a brand-new Solo button (which automatically silences non-soloed lanes while active), and both its currently playing sound (`active_timbre`) and its next-bar preloaded sound (`queued_timbre`).
- **Exact Musical Scaling Curves**: Turning a physical MIDI knob from `0` to `127` scales note density exponentially from `0.1x` to `5.0x` (`0.1 * 50^(cc / 127)`). For rhythmic feel (`swing` and `humanize`), standard 7-bit MIDI has an odd number of values (`0` to `127`), meaning `64 / 127` is slightly off-center (`0.5039...`). Our feel curve uses piecewise linear interpolation so the physical center detent at `CC 64` produces **exact** `0.5` (50% of the authored swing and humanize).
- **4,096-Form & 4,096-Timbre Reservoirs**: Tracks the active musical form across a 4,096-entry bank (with history and a 4-form lookahead prefetch window) and 4,096 full 8-voice timbre sets (with automatic refill at `1,024` remaining sets and FNV-1a hash placement for saved snapshots).
- **Dual Operating Modes (Native nanoKONTROL2 + MIDImix Emulation)**: Supports both a direct, purpose-built Korg `nanoKONTROL2` performance layout (where `Marker Set CC 60` captures an atomic preset snapshot in a single button press) and a backwards-compatible `MIDImix` Emulation mode that translates `nanoKONTROL2` actions into factory Akai `MIDI Mix` Notes and CCs—including synthesizing the full 32-CC `Send All` burst when `Marker Set CC 60` is pressed.

Plain-English Concept	Rust Symbol (‘src/op8_model.rs‘)	Node.js Symbol (‘js/op8-nanokontrol2-mapper.js‘)	Technical Behavior
8 Percussion Voice Lanes	`FAMILY_ORDER` / `Op8VoiceLane`	`FAMILY_ORDER` / `model.lanes`	Models `kick`, `snare`, `hat_closed`, `hat_open`, `clap`, `tom`, `rim`, `cymbal` (plus Konduktiva aliases)
Exponential Density Ladder	`cc_to_density` / `density_to_cc`	`mapMidiCCToDensity` / `mapDensityToMidiCC`	Maps `CC 0..127` to `0.1 * 50^(cc / 127)` (`0.1x` to `5.0x`)
Exact Midpoint Feel Curve	`cc_to_feel_scale` / `cc_to_midimix_feel_scale`	`mapMidiCCToFeelScale` / `mapMidiCCToMidimixFeelScale`	Guarantees `CC 64 == 0.5` exact midpoint for `swing` and `humanize`
Solo & Mute Audibility	`Op8PerformanceModel::is_lane_audible`	`Op8PerformanceModel.prototype.isLaneAudible`	Honors Mute state and gives Solo priority over non-soloed lanes
4,096-Entry Form Bank	`Op8FormBankState`	`model.formBank`	Circular `0..4096` navigation with history and 4-form prefetch window
4,096-Entry Timbre Bank	`Op8TimbreBankState` / `Op8PresetSnapshot`	`model.timbreBank` / `timbrePoolIndex`	Refills at `1024` and hashes saved snapshots via 32-bit FNV-1a
Native nanoKONTROL2 Mapper	`Op8PerformanceModel::apply_nanokontrol2_cc`	`Op8PerformanceModel.prototype.applyNanoKontrol2Cc`	Maps all 8 track strips and 11 transport buttons to `Op8PerformanceAction`
MIDImix Emulation Translator	`translate_nanokontrol2_to_midimix`	`translateNanoKontrol2ToMidimix`	Emits factory MIDImix Notes (`1..27`), CCs (`17..61`), and 32-CC `Send All` burst

Native nanoKONTROL2 vs. MIDImix Emulation Mapping Reference

Physical nanoKONTROL2 Control	Native nanoKONTROL2 CC	8opPerc Native Action	Emulated MIDImix Message (Channel 1)
Track Knobs 1..8	`CC 16..23`	`SetLaneDensity` (`0.1 .. 5.0`)	Row 2 Voice Density `CC 17, 21, 25, 29, 47, 51, 55, 59`
Track Sliders 1..8	`CC 0..7`	`SetLaneGain` (`0..127` / `0.0..1.0`)	Strip Faders `CC 19, 23, 27, 31, 49, 53, 57, 61`
Track Solo Buttons 1..8	`CC 32..39`	`ToggleLaneSolo` (with Solo priority mask)	Solo Buttons `Note 2, 5, 8, 11, 14, 17, 20, 23`
Track Mute Buttons 1..8	`CC 48..55`	`ToggleLaneMute` (with Audible mask)	Mute Buttons `Note 1, 4, 7, 10, 13, 16, 19, 22`
Track Rec Buttons 1..8	`CC 64..71`	`RerollLaneTimbre` (queues next-bar patch)	Rec Arm Buttons `Note 3, 6, 9, 12, 15, 18, 21, 24`
Track Left / Right	`CC 58 / CC 59`	`NavigateForm` (`Previous` / `Next` in 4,096 bank)	Bank Left `Note 25` / Bank Right `Note 26`
Cycle Button	`CC 46`	`AdvanceTimbreSet` (next 8-voice timbre set)	Solo Master `Note 27`
Marker Set Button	`CC 60`	`CapturePresetPoolSnapshot` (single-button atomic save)	Full 32-CC `Send All` burst (`CC 16..31` and `CC 46..61`)
Marker Left / Right	`CC 61 / CC 62`	`StepGlobalDensity` (−8 / +8 CC step)	Global Density Fader `CC 61`
Rewind / Forward	`CC 43 / CC 44`	`StepGlobalFeel` (−8 / +8 CC step around `CC 64`)	Global Feel Knob `CC 58`
Play / Stop / Record	`CC 41 / CC 42 / CC 45`	`TransportPlay`, `TransportStop`, `TransportRecord`	Native transport control

Tests

QA

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**Compilation & Test Verification**: `CARGO_BUILD_JOBS=2 cargo test` passed 38/38 tests with zero warnings or errors; `npm test` passed 9/9 Node.js tests (`47` total automated tests).

**Audio & GUI Safety Policy**: Verified purely via deterministic in-memory state transitions; no GUI window or live `8opPerc` audio synthesis process was launched.
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Files touched

- ‘src/op8_model.rs‘ (Created — 792 lines: ‘Op8VoiceLane‘, ‘Op8RhythmicFeel‘, ‘Op8FormBankState‘, ‘Op8TimbreBankState‘, ‘Op8PresetSnapshot‘, ‘Op8PerformanceModel‘, ‘Op8PerformanceAction‘, ‘EmulatedMidiMessage‘, ‘cc_to_density‘, ‘cc_to_feel_scale‘, and 5 unit tests)
- ‘src/lib.rs‘ (Modified — exported ‘pub mod op8_model;‘)
- ‘js/op8-nanokontrol2-mapper.js‘ (Created — 514 lines: ‘Op8PerformanceModel‘, ‘mapMidiCC-

ToDensity‘, ‘mapMidiCCToFeelScale‘, ‘mapMidiCCToMidimixFeelScale‘, ‘timbrePoolIndex‘, ‘applyNanoKontrol2Cc‘, and ‘translateNanoKontrol2ToMidimix‘)

- ‘js/index.js‘ (Modified — re-exported ‘./op8-nanokontrol2-mapper.js‘)
- ‘js/nkeditor-sdk.test.js‘ (Modified — added Epoch 8 test suite for ‘Op8PerformanceModel‘ and scaling functions)

Test coverage (47 total)

QA verdict

PASS

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